

QUESTIONS

10 / 10 completed

Sparse Matrices

SPARSE MATRICES

Represent two sparse matrices (mostly zeros) using arrays of triples {row, col, value}. Read dimensions and non-zero terms for two matrices. Add them, store result in sparse form (sorted by row then col), and display dense matrix form. Max 10x10; ignore zeros.

Sample Input (Matrix 1):

3 3 3
0 1 5
1 0 8
2 2 2

Sample Input (Matrix 2):

3 3 2
0 1 3
2 2 7

Sample Output (Sum Dense):

0 8 0
8 0 0
0 0 9

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main() {
3     int a[10][10] = {0}, b[10][10] = {0};
4     int r, c, n, i, j, x, y, value;
5     scanf("%d %d %d", &r, &c, &n);
6     for(i = 0; i < n; i++) {
7         scanf("%d %d %d", &x, &y, &value);
8         a[x][y] = value;
9     }
10    scanf("%d %d %d", &r, &c, &n);
11    for(i = 0; i < n; i++) {
12        scanf("%d %d %d", &x, &y, &value);
13        b[x][y] = value;
14    }
15    printf("Sum Dense:\n");
16    for(i = 0; i < r; i++) {
17        for(j = 0; j < c; j++) {
18            printf("%d ", a[i][j] + b[i][j]);
19        }
20        printf("\n");
21    }
```

OUTPUT

```
Sum Dense:
0 8 0
8 0 0
0 0 9
```

INPUT

3 3 3
0 1 5